that it needed electricity, steel, land and other resources to build more factories and produce more paper clips, and realising that humans are unlikely to give up these resources, the superintelligent computer eliminated all humans in its single-minded pursuit of its given goal.[26] Bostrom's point was that the problem with computers isn't that they are particularly evil but that they are particularly powerful. And the more powerful the computer, the more careful we need to be about defining its goal in a way that precisely aligns with our ultimate goals. If we define a misaligned goal to a pocket calculator, the consequences are trivial. But if we define a misaligned goal to a superintelligent machine, the consequences could be dystopian.

The paper-clip thought experiment may sound outlandish and utterly disconnected from reality. But if Silicon Valley managers had paid attention when Bostrom published it in 2014, perhaps they would have been more careful before instructing their algorithms to 'maximise user engagement'. The Facebook and YouTube algorithms behaved exactly like Bostrom's imaginary algorithm. When told to maximise paper-clip production, the algorithm sought to convert the entire physical universe into paper clips, even if it meant destroying human civilisation. When told to maximise user engagement, the Facebook and YouTube algorithms sought to convert the entire social universe into user engagement, even if it meant doing harm to the social fabric of Myanmar, Brazil and many other countries.

Bostrom's thought experiment highlights a second reason why the alignment problem is more urgent in the case of computers. Because they are inorganic entities, they are likely to adopt strategies that would never occur to any human and that we are therefore ill-equipped to foresee and forestall. Here's one example: In 2016, Dario Amodei was working on a project called Universe, trying to develop a general-purpose AI that could play hundreds of different computer games. The AI competed well in various car races, so Amodei next tried it on a boat race. Inexplicably, the AI steered its boat right into a harbour and then sailed in endless circles in and out of the harbour.

It took Amodei considerable time to understand what went wrong. The problem occurred because initially Amodei wasn't sure how to tell the AI that its goal was to 'win the race'. 'Winning' is an unclear concept to an algorithm. Translating 'win the race' into computer language would have required Amodei to formalise complex concepts like track position and